ProbOS — Month 3, Week 14
Injection Molding Process Qualification (Research Gate First)

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Week 14 goal

Add injection molding process qualification as a new domain model, reusing the existing, validated engine (`Distribution`/ `Model` ABCs, `MonteCarloEngine`, `SobolSensitivity`, the now-fixed reproducible seeding, C++ `LogNormal`/ `Uniform` classes, PDSL v0.2) — explicitly NOT a new architecture, a new model class on infrastructure already at 408/408 green tests.

Decision, stated up front: build the real model first, with a genuine research gate (Day 1, before any model code), THEN add it as a fourth tab to the live demo — not assuming the originating pitch document's claim that "public literature on PP/PC process windows" exists without verifying it directly.

Day 1 – Research gate (before any model code)

- Search for a real, citable, published injection molding process-window dataset (PP or PC material) with real reported process parameters and real reported out-of-spec/defect outcomes — the exact discipline already used for `PharmaStabilityModel` (Gonzalez-Gonzalez et al. 2023)
- **Explicit decision gate:** if real, usable data is found, proceed to Day 2 as planned below. If not found, document the gap honestly and either narrow the model's scope to what real data DOES support, or defer the week's remaining time to another already-planned item

Day 2 – InjectionMoldingProcessModel (contingent on Day 1)

- Same architectural pattern as `BatteryModel2Cell`: state variables (cavity pressure, melt temperature, fill time, cooling time, part dimensional outcome), parameters with real uncertainty informed by Day 1's findings (resin lot viscosity — reuses the existing C++ `LogNormal` class directly; ambient humidity and machine repeatability as `Normal`; wall- thickness tolerance from mold wear)
- Output: probability of an out-of-spec part as a function of the process window — directly analogous to the battery model's thermal-runaway probability output

Day 3 – Monte Carlo + Sobol sensitivity

- Wire the real `MonteCarloEngine` and `SobolSensitivity` (now genuinely reproducible, per this session's own seeding-bug fix and its permanent regression test) to the new model

- The actual sellable insight: “of N process parameters, which 2–3 are actually driving scrap rate”

Day 4 – Validation against the real Day 1 dataset

- Validate against whatever real dataset Day 1 found, with an honestly stated tolerance and methodology strength

Day 5 – PDSL spec for process windows

- A real PDSL v0.2 example letting a process engineer define a process window declaratively

Day 6 – Fourth demo tab

- Add Injection Molding as a fourth tab to the live demo, following the EXACT same discipline already established: real pre-computed kernel data, no JS toy-formula reimplementation, real interpretation bands, real Sobol breakdown, hover tooltips

Day 7 – Audit, documentation, retrospective

- `pre_week_audit.sh`, full test suite, CI, retrospective

Business framing (for reference, not this week’s engineering scope)

Open-core (Apache 2.0) for the model and Sobol analysis; a paid audit-trail tier reframes the earlier MoldMind + ValidTrail ideas as one feature on existing infrastructure. Outreach is explicitly OUT of this week’s scope.

Exit criteria

A genuinely validated (or honestly gap-documented) injection molding domain model, wired to the real Sobol/Monte Carlo engine, with a working PDSL example and a fourth live demo tab built on real data.

What was actually accomplished (retrospective)

- A third real, validated domain (`InjectionMoldingProcessModel`) added on existing, unmodified infrastructure – zero new architecture, matching the week’s own stated goal.
- Day 1’s research gate found two real, independent sources: Kavade & Kadam (2012) (real physical experiments, the primary validation target) and Moayyedian et al. (2021) (FEA simulation, used only as a structural cross-check) – exactly the “don’t assume a pitch document’s citation claim without verifying it” discipline stated up front in this week’s own plan.
- The model’s own real Sobol sensitivity ranking reproduced ALL FIVE parameters in the exact same relative order as the source paper’s independently-measured real ANOVA ranking, now covered by a permanent regression test.

- An honest architectural difference was documented, not hidden: unlike `BatteryModel2Cell` or `PharmaStabilityModel`, injection molding is a genuinely static process (confirmed idempotent) – the demo tab accordingly skips the time-series chart rather than faking a flat placeholder trajectory.

Real bugs found and fixed during this week

1. **PDSL drift semantics (Day 5):** the first PDSL source wrote the drift expression as if it directly SET the new state value; PDSL's actual, established semantics are Euler-integration style (`new_state = state + dt*drift`), matching Battery/Pharma. Confirmed via cross-validation against the hand-written Python model: the mean was off by exactly the nominal weight itself (96.826g), diagnosed via direct computation before fixing.

2. **Readout grid collapse (found via direct user review):** hiding grid child elements with `display:none` does not collapse a CSS grid container's own column tracks, leaving a large empty blank space. Fixed by dynamically changing `grid-template-columns` on the container itself.

3. **Stale Sobol/PDSL panel titles (found via direct user review):** both panels were hard-coded to always describe "battery," even while displaying injection molding's own real data – confirmed as a real correctness bug, not just a cosmetic issue, and fixed with genuine per-tab dynamic content.

What went right

1. Direct user review of live screenshots caught two real UI bugs that automated checks (DOM-ID matching, balance checks) could not have caught, since both were about what actually RENDERS, not structural correctness.

2. A suspected third bug (histogram bin coloring) was correctly resolved as NOT a bug after full bin-by-bin computation – avoided "fixing" correctly-working code based on an incomplete percentile-level check.

Numbers at Week 14 close

- Python tests: 419/419 passing
- mypy strict: 0 errors: ruff: 0 warnings
- Real Sobol structural validation: exact match to the source paper's ANOVA ranking, all 5 parameters
- Fourth live demo tab, real data, zero known workarounds